

Let's begin our analysis of this circuit, assuming that the output is high ($= V_{DD}$) and the input is low ($= 0$ V). Figure 18.4 shows the bottom portion of the Schmitt trigger used in calculating the upper switching point voltage, V_{SPH} . MOSFETs M1 and M2 are off, with $V_{in} = 0$ V while M3 is on. The source of M3 floats to $V_{DD} - V_{THN}$, or approximately 4 V for $V_{DD} = 5$ V. We can label this potential V_x , as shown in the figure.

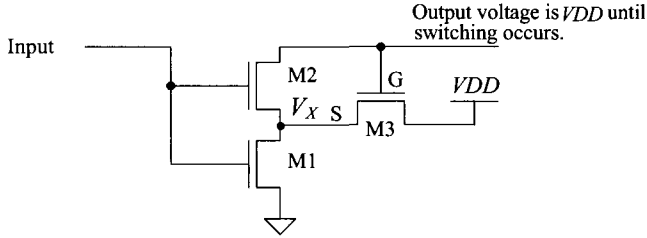


Figure 18.4 Portion of the Schmitt trigger schematic used to calculate upper switching point voltage.

With V_{in} less than the threshold voltage of M1, V_x remains at $V_{DD} - V_{THN3}$. As V_{in} is increased further, M1 begins to turn on and the voltage, V_x , starts to fall toward ground. The high switching point voltage is defined when

$$V_{in} = V_{SPH} = V_{THN2} + V_x \quad (18.2)$$

or when M2 starts to turn on. As M2 starts to turn on, the output starts to move toward ground, causing M3 to start turning off. This in turn causes V_x to fall further, turning M2 on even more. This continues until M3 is totally off and M2 and M1 are on. This positive feedback causes the switching point voltage to be very well defined.

When Eq. (18.2) is valid, the currents flowing in M1 and M3 are essentially the same. Equating these currents gives

$$\frac{\beta_1}{2}(V_{SPH} - V_{THN})^2 = \frac{\beta_3}{2}(V_{DD} - V_x - V_{THN3})^2 \quad (18.3)$$

Since the sources of M2 and M3 are tied together, $V_{THN2} = V_{THN3}$, the increase in the threshold voltages from the body effect is the same for each MOSFET. The combination of Eqs. (18.2) and (18.3) yields

$$\frac{\beta_1}{\beta_3} = \frac{W_1 L_3}{L_1 W_3} = \left[\frac{V_{DD} - V_{SPH}}{V_{SPH} - V_{THN}} \right]^2 \quad (18.4)$$

The threshold voltage of M1, given by V_{THN} in this equation, is the zero body bias threshold voltage ($= 0.8$ V in our long-channel CMOS process and 0.25 V in the short-channel process). Given a specific upper switching point voltage, the ratio of the MOSFET transconductors is determined by solving this equation. A general design rule for selecting the size of M2, that is, β_2 , is to require that

$$\beta_2 \geq \beta_1 \text{ or } \beta_3 \quad (18.5)$$

since M2 is used as a switch.

A similar analysis can be used to determine the lower switching point voltage, V_{SPL} , resulting in the following design equation:

$$\frac{\beta_5}{\beta_6} = \frac{W_5 L_6}{L_5 W_6} = \left[\frac{V_{SPL}}{V_{DD} - V_{SPL} - V_{THP}} \right]^2 \quad (18.6)$$

The following example illustrates the design procedure for a Schmitt trigger.

Example 18.1

Design and simulate a Schmitt trigger using the short-channel CMOS process with $V_{SPL} = 400$ mV and $V_{SPH} = 700$ mV.

We begin by solving Eqs. (18.4) and (18.6) for the transconductance ratios. For the upper switching point voltage,

$$\frac{W_1 L_3}{W_3 L_1} = \left[\frac{1 - 0.7}{0.7 - 0.25} \right]^2 = 0.444 \rightarrow L_1 = L_3 = 1 \text{ and } W_1 = 10, W_3 = 22.5$$

and for the lower switching point voltage,

$$\frac{W_5 L_6}{W_6 L_5} = \left[\frac{0.4}{1 - 0.4 - 0.25} \right]^2 = 1.3 \rightarrow L_5 = L_6 = 1 \text{ and } W_6 = 20, W_5 = 26$$

M2 is set to 10/1 and M4 is set to 20/1. Simulation results are seen in Fig. 18.5. This figure reveals the benefit of using a Schmitt trigger, namely, it allows slow moving inputs to be made into good solid logic high and low values. In a practical circuit, connecting the ramp-shaped input seen in Fig. 18.5 to an inverter would produce oscillations in the inverter's output (because of noise on the ramp). ■

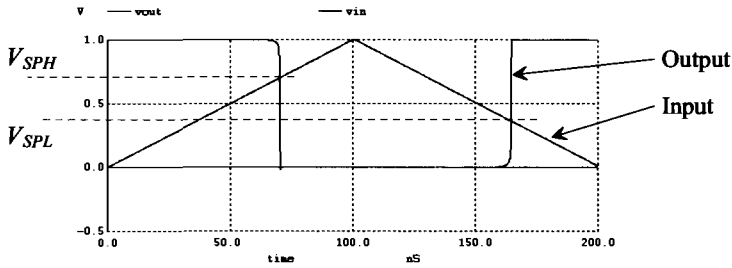


Figure 18.5 The input and output of the Schmitt trigger designed in Ex. 18.1.

Switching Characteristics

The propagation delays of the Schmitt trigger can be calculated in much the same way as the inverter of Ch. 11. Defining equivalent digital resistances for M1, M2, M4, and M5 as R_{n1} , R_{n2} , R_{p4} , and R_{p5} , respectively, gives a high-to-low propagation delay-time, neglecting the Schmitt trigger output capacitance, of

$$t_{PHL} = 0.7 \cdot (R_{n1} + R_{n2}) \cdot C_{load} \quad (18.7)$$

and

$$t_{PLH} = 0.7 \cdot (R_{p4} + R_{p5}) \cdot C_{load} \quad (18.8)$$

18.1.2 Applications of the Schmitt Trigger

Consider the waveform shown in Fig. 18.6. A pulse with ringing is a common voltage waveform encountered in buses or lines interconnecting systems. If this voltage is applied directly to a logic gate or inverter input with a V_{SP} of 0.5 V, the output of the gate will vary with the period of the ringing on top of the pulse. Using a Schmitt trigger with properly designed switching points can eliminate this problem.

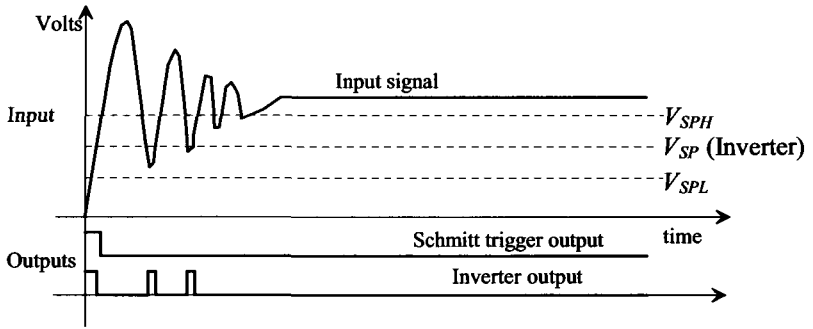


Figure 18.6 Applying a Schmitt trigger to clean up an interconnecting signal.

The Schmitt trigger can also be used as an oscillator (Fig. 18.7). The delay-time in charging and discharging the capacitor sets the oscillation frequency. At the moment in time when the output of the Schmitt trigger switches low, the voltage across the capacitor is V_{SPH} . The capacitor will start to discharge toward ground. The voltage across the capacitor is given by

$$V_c(t) = V_{SPH} \cdot e^{-t/RC} \quad (18.9)$$

At the time when $V_c(t) = V_{SPL}$, the output of the Schmitt trigger changes state. This time is given by solving Eq. (18.9) by

$$t_1 = RC \cdot \ln \frac{V_{SPH}}{V_{SPL}} \quad (18.10)$$

A similar analysis for the case when the capacitor is charged from V_{SPL} to V_{SPH} gives

$$V_c(t) = V_{SPL} + (V_{DD} - V_{SPL}) \left(1 - e^{-\frac{t}{RC}}\right) \quad (18.11)$$

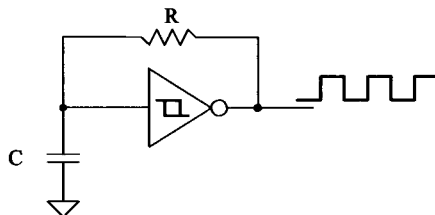


Figure 18.7 Oscillator design using a Schmitt trigger.

and

$$t_2 = RC \cdot \ln \frac{V_{DD} - V_{SPL}}{V_{DD} - V_{SPH}} \quad (18.12)$$

The oscillation frequency, neglecting the intrinsic delay of the Schmitt trigger, is given by

$$f_{osc} = \frac{1}{t_1 + t_2} \quad (18.13)$$

The capacitance used in these equations is the sum of the input capacitance of the Schmitt trigger and any external capacitance.

An alternative oscillator using the Schmitt trigger is shown in Fig. 18.8. Here the MOSFETs M1 and M4 behave as current sources (see Ch. 20) mirroring the current in M5 and M6. When the output of the oscillator is low, M3 is on and M2 is off. This allows the constant current from M4 to charge C . When the voltage across C reaches V_{SPH} , the output of the Schmitt trigger swings low. This causes the output of the oscillator to go high and allows the constant current from M1 to discharge C . When C is discharged down to V_{SPL} , the Schmitt trigger changes states. This series of events continues, generating the square wave output.

If we label the drain currents of M1 and M4 as I_{D1} and I_{D4} , we can estimate the time it takes the capacitor to charge from V_{SPL} to V_{SPH} as

$$t_1 = C \cdot \frac{V_{SPH} - V_{SPL}}{I_{D4}} \quad (18.14)$$

and the time it takes to charge from V_{SPH} to V_{SPL} is

$$t_2 = C \cdot \frac{V_{SPH} - V_{SPL}}{I_{D1}} \quad (18.15)$$

The period of the oscillation frequency is, as before, the sum of t_1 and t_2 .

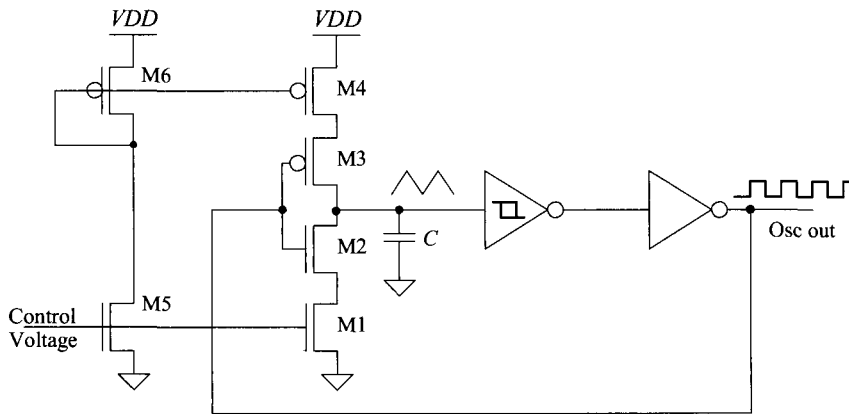


Figure 18.8 Voltage-controlled oscillator using Schmitt trigger and current sources. MOSFETs M2 and M3 are used as switches.

This type of oscillator is termed a voltage-controlled oscillator (VCO) since the output frequency can be controlled by an external voltage. The currents I_{D1} and I_{D4} , (Fig. 18.8) are directly controlled by the control voltage. As we will see in Ch. 20, the current in M5 is mirrored in M1, M4, and M6, with an appropriate scaling factor dependent on the size of the transistors.

18.2 Multivibrator Circuits

Multivibrator circuits (Fig. 18.9) are circuits that employ positive feedback. The name “multivibrator” is a vestige from early-time electronics development (prior to the ubiquitous term “digital”) where the circuits’ outputs vibrate between two states. There are three types of multivibrators: astable, bistable, and monostable. Astable multivibrator circuits are unstable in either output (high or low) state. The oscillators that we have discussed are examples of astable multivibrators. The bistable multivibrator is stable in either the high or low state. Flip-flops and latches are examples of the bistable multivibrator. Monostable multivibrators are stable in a single state. Monostable multivibrators are also called one-shots. In this section we discuss the monostable and astable multivibrators. We distinguish the material in this section from the other material in the book by using resistor-capacitor time constants to set time intervals.

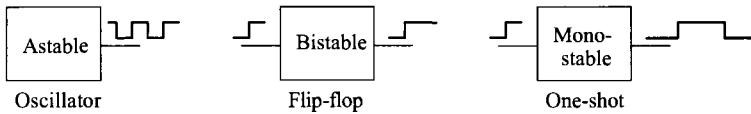


Figure 18.9 Multivibrator circuits.

18.2.1 The Monostable Multivibrator

A CMOS implementation of the monostable multivibrator is shown in Fig. 18.10. Under normal conditions, V_{in} is low and the output of the NOR gate, V_1 , is high. The voltage V_2 is pulled high through the resistor, and the output of the inverter, V_3 , is a low. Upon application of a trigger pulse, that is, V_{in} going high, both V_1 and V_2 drop to zero volts and the output of the inverter, which is also the output of the monostable, goes high. This output is fed back to the input of the NOR gate holding V_1 at ground potential.

After triggering, the potential V_2 will start to increase because C is charged through R . The potential across the capacitor after triggering takes place is given by

$$V_c(t) = V_2(t) - \overbrace{V_1(t)}^{=0} = VDD \cdot (1 - e^{-\frac{t}{RC}}) \quad (18.16)$$

If we assume that the V_{SP} of the inverter is $VDD/2$, then the time it takes for the capacitor to charge to V_{SP} is given by

$$t = RC \cdot \ln \frac{VDD}{VDD - V_{SP}} = RC \cdot \ln(2) \approx 0.7RC \quad (18.17)$$

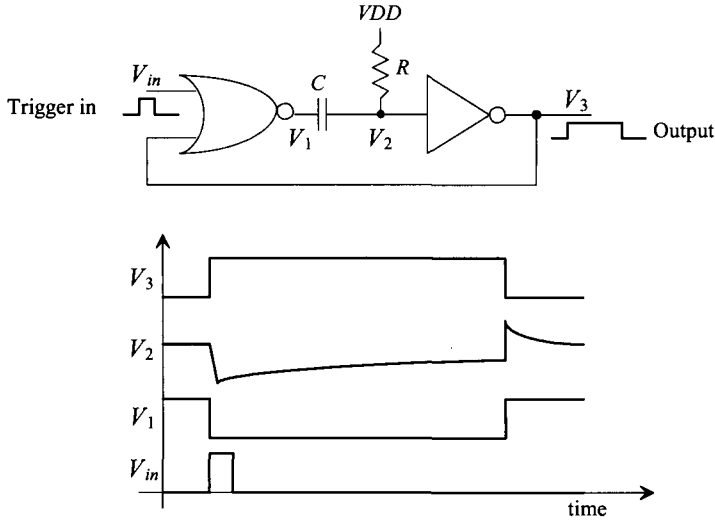


Figure 18.10 Operation of the monostable multivibrator.

This time also defines the output pulse width, neglecting gate delays, since the inverter switches low. The inverter output, V_3 , going low causes V_1 to go back to V_{DD} and V_2 to go to $V_{DD} + V_{DD}/2$. If the resistor or capacitor is bonded out (connected to the output pads), the ESD diodes may keep V_2 from going much above $V_{DD} + 0.7$. The time it takes V_2 to decay back down to V_{DD} limits the rate at which the one-shot can be retriggered. Also note that the trigger input can be longer than the output pulse width. Longer output pulse widths may cause V_2 to go as high as 10 V as well as limit the maximum trigger rate.

18.2.2 The Astable Multivibrator

An example of an astable multivibrator is shown in Fig. 18.11. This circuit has no stable state and thus oscillates. To analyze the behavior of this multivibrator, let's begin by assuming that the output, V_3 , has just switched high. The output going high causes V_1 to go high (to $V_{DD} + V_{SP1}$), forcing V_2 low. The voltage across the capacitor after this switching takes place is given by

$$V_c(t) = V_1(t) - \overbrace{V_3(t)}^{= V_{DD}} = (V_{DD} + V_{SP1}) \cdot e^{-\frac{t}{RC}} - V_{DD} \quad (18.18)$$

The output of the astable will go low, $V_3 = 0$, when $V_1 = V_{SP1}$. Substituting this condition into the previous equation gives the time the output is high (or low) and is given by

$$t_1 = RC \cdot \ln \frac{V_{DD} + V_{SP1}}{V_{SP1}} = t_2 \quad (18.19)$$

If $V_{SP1} = V_{DD}/2$, then

$$t_1 = t_2 = 1.1RC \quad (18.20)$$

and the frequency of oscillation is

$$f_{osc} = \frac{1}{t_1 + t_2} = \frac{1}{2.2RC} \quad (18.21)$$

Again, if the resistor and capacitor are bonded out, the ESD diodes on the pads will limit the voltage swing of V_1 .

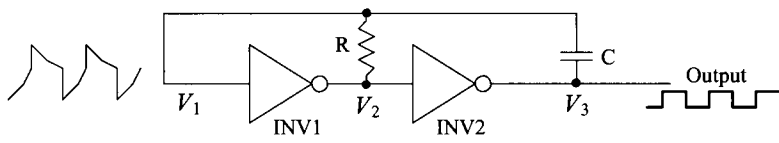


Figure 18.11 An astable multivibrator.

18.3 Input Buffers

Input buffers are circuits that take a chip's input signal, with imperfections such as slow rise and fall times, and convert it into a clean digital signal for use on-chip. If the buffer doesn't "slice" the data in the correct position, timing errors can occur. For example, consider the waveforms seen in Fig. 18.12. If the input signal is sliced too high or too low, the output signal's width is incorrect. In high-speed systems this reduces the timing budget in the system and can result in errors.

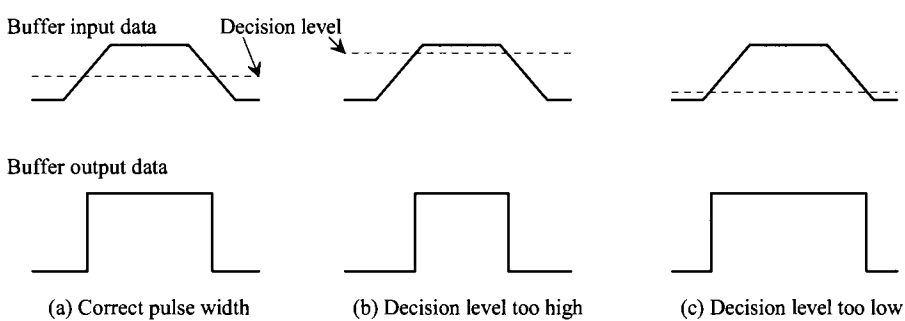


Figure 18.12 Timing errors in regenerating digital data.

18.3.1 Basic Circuits

If not careful, simple circuits can introduce unwanted pulse width variations because of differences in rise and fall times. Consider the inverter circuit seen in Fig. 18.13. The switching point voltage of the inverter, because of process, V_{DD} (voltage), or temperature variations (PVT) will move around. Further, differences in the PMOS and the NMOS resistance will affect the rise and fall times of the inverter's output signal. The output of the first inverter, in Fig. 18.13, sees a much larger load capacitance than the second inverter. The delay time from the input going high to the output going high is roughly

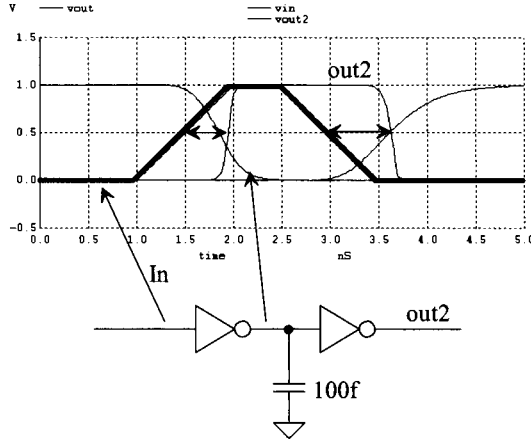


Figure 18.13 How skew is introduced into a high-speed signal.

400-ps. However, the delay time between the input and the output going low is 600 ps. As the input signal travels through the digital system, the time skew added to the input signal can add up and result in timing errors (the data is received too early or too late at different points in the system). Note that had we matched the loading of each inverter, the propagation delays would be more similar, Fig. 18.14. When the input signal goes high, the output of the first inverter goes low (t_{PHL}) and the output of the second inverter goes high (t_{PLH}). The total delay is the sum of $t_{PHL} + t_{PLH}$. However, when the input signal goes low, the output of the first inverter goes high (t_{PLH}) and the output of the second inverter goes low (t_{PHL}), resulting in the same sum ($t_{PHL} + t_{PLH}$). *If possible, make the number of t_{PHL} delays in series with a high-speed signal even and equal to the number of t_{PLH} delays.*

Of course, to better equalize the delays, the second inverter in Fig. 18.14 would need an additional bit of loading (an inverter connected to its output). Also, the input signal should transition faster in an attempt to match the transition times of the inverters.

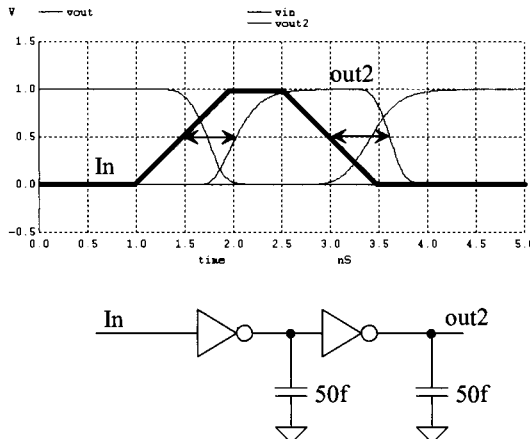


Figure 18.14 How equalizing delays can be used to reduce skew.

Skew in Logic Gates

Consider the NAND gate in Fig. 18.15. A different delay will be added to an input signal depending on which input of the NAND gate is used. As seen in the figure, the *A* input, with the *B* input high, propagates to the output slightly quicker than the *B* input (with the *A* input high). For the NAND gate, this difference in propagation delays is only a factor for the NMOS devices (because they are in series with the output of the gate).

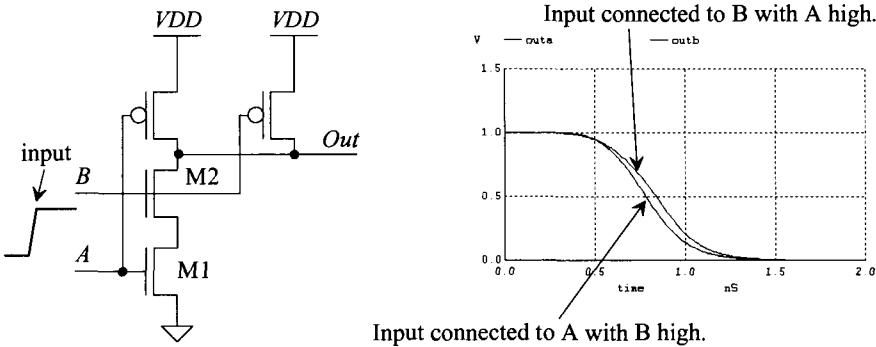


Figure 18.15 The skew introduced by using different inputs.

Figure 18.16 shows how two NAND gates can be used in parallel to eliminate the differences in the propagation delay between inputs. The series NMOS are arranged so that no matter which input is toggled the output changes at the same rate. Note that two of the PMOS devices can be eliminated to simplify the circuit without affecting the propagation delays.

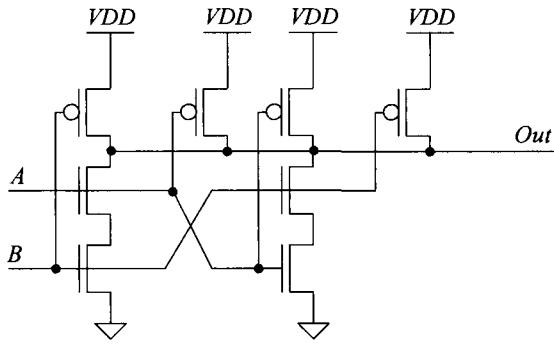


Figure 18.16 Using two NAND gates in parallel to reduce input-dependent skew.

The previous discussion neglected the effects of the gate's switching point voltage. If the input signals are not transitioning to full logic levels (V_{DD} and ground) or the rise and fall times are not fast (compared to the gate delay time), then the point where the input circuit switches is critical. This leads us to our next topic.

18.3.2 Differential Circuits

In order to precisely slice input data, as seen in Fig. 18.12, a reference voltage may be transmitted, on a different signal path, along with the data. Alternatively, the data may be transmitted differentially (an input and its complement). In either case a differential amplifier is needed, Fig. 18.17. A differential amplifier input buffer (which we'll simply call an input buffer from this point on) amplifies the difference between the two inputs. In the simplest case one input to the input buffer is a DC voltage, say 0.5 V (V_{inm} in Fig. 18.17). When the other input (V_{inp}) goes above 0.5 V, the output of the buffer changes states (goes from a low to a high) or

$$V_{inp} > V_{inm} \rightarrow \text{out} = "1"$$
(18.22)

and

$$V_{inp} < V_{inm} \rightarrow \text{out} = "0"$$
(18.23)

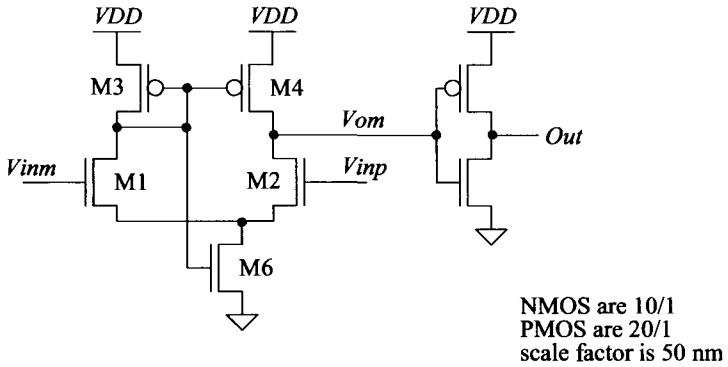


Figure 18.17 An (n-flavor) input buffer for high-speed digital design.

The diff-amp in Fig. 18.17 is based on the topologies seen in Ch. 22. This circuit is *self-biased* because no external references are used to set the current in the circuit (the gate of M6 is tied up to the gate of M3). When V_{inp} is larger than V_{inm} , the current in M2 is larger than the current in M1 ($V_{GS2} > V_{GS1}$). The current in M1 flows through M3 and is mirrored by M4 (and so M4's current is less than M2's current). This causes the diff-amp's output, V_{om} , to go towards ground (until the current in M2 equals the current in M4) and the output of the inverter, Out , to go high. Note that the gain from the V_{inp} input to the output of the circuit is larger than the gain from V_{inm} to the output (M3, being diode-connected, is a lower resistance than M4). It is generally a good idea to connect the reference voltage to the V_{inm} input.

The DC simulation results for the buffer in Fig. 18.17 are shown in Fig. 18.18. The x-axis is the V_{inp} input swept from 0 to 1 V. The V_{inm} input is held from 0 to 1 V in 200 mV steps. When V_{inm} is held at 400 mV and V_{inp} goes above 400 mV, the output changes from 0 to 1 (although there is a small offset which necessitates that V_{inp} go to approximately 415 mV before the output transitions).

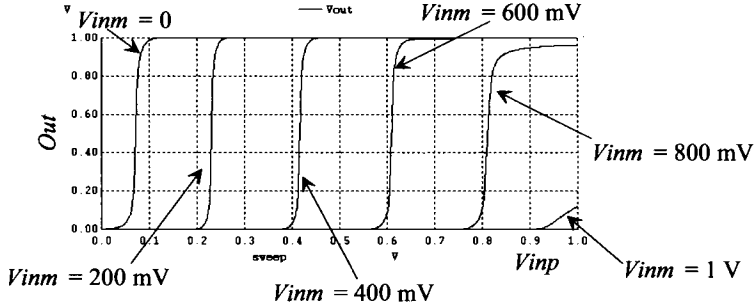


Figure 18.18 Simulating the DC behavior of the buffer in Fig. 18.17.

Transient Response

An example transient response for the buffer in Fig. 18.17 is seen in Fig. 18.19. A very small increase in V_{inp} above V_{inm} is required to make the output of the buffer switch states. We have several practical questions that should be answered with variations of this simulation. For example, what are the minimum and maximum values of V_{inm} allowed. Next, why are the delays between V_{inp} going high and going low different? Is it because of the offset seen in Fig. 18.18 (the output doesn't precisely switch at a V_{inm} 400 mV)? Let's provide some discussion concerning these concerns.

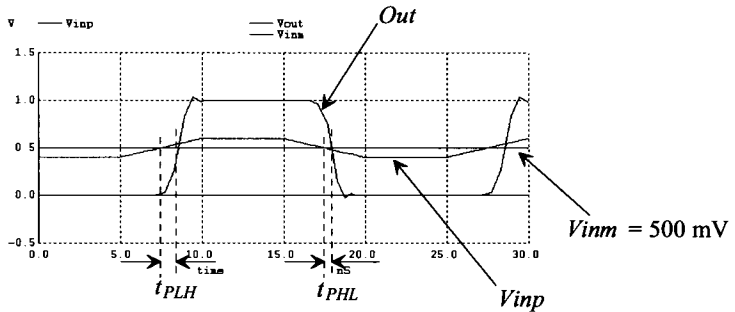


Figure 18.19 Transient response of the input buffer in Fig. 18.17 driving a 50 fF load.

Looking at Fig. 18.17, we can see that if the inputs fall below V_{THN} ($= 250$ mV here), then the circuit won't work very quickly (the MOSFETs move into the subthreshold region). So we would expect the propagation delays to increase. Indeed, as seen in Fig. 18.20, the delays go up. Ideally, the delay of the buffer is independent of power supply voltage, temperature, or input signal amplitudes (or pulse shape). To get better performance for lower input level signals, we might use the PMOS version of the buffer in Fig. 18.17, as seen in Fig. 18.21. Resimulating this buffer with the signals seen in Fig. 18.20 gives the results seen in Fig. 18.22. The delays are considerably better, however, there is an offset that appears rather large (because the output changes at the same time as V_{inp} going past V_{inm} , indicating that an offset is present). To avoid this